

Priority-Conditioned Retention: A Controllable Middle-Layer Memory Circuit in Pretrained Mamba

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Abstract. State-space models (SSMs) such as Mamba compress an entire history into a fixed-size continuous hidden state, achieving linear-time inference at the cost of associative recall: unlike attention, they cannot losslessly retrieve an arbitrary past token. Prior work attributes this to a capacity bound but leaves open whether the practical failure is a *storage* limit or a *control* limit. We show that a **pretrained** Mamba-790m already contains a **controllable, spatially localized memory circuit**, accessible at inference with no fine-tuning. By scaling the pre-softplus input to the Δ gate at positions downstream of a target, we reduce the effective decay applied to that target and recover associative recall on a capacity-pressured multi-query associative-recall (MQAR) task ($0.53 \rightarrow 0.73$ at the optimal operating point). The effect is dose-dependent, target-specific, and bidirectional, and it is **localized to layers 16–23**. Restricting the intervention to these 8 of 48 layers recovers most of the recall gain at **near-zero language-modeling cost** (**perplexity** $\times 1.07$), whereas the all-layer intervention inflates perplexity $\times 1.81$. We frame this as the continuous-state analogue of importance-aware KV-cache compression and report an honest boundary: a literal write-quantization analogue does *not* transfer, because the superimposed state has no per-item slots to reallocate.

1 Introduction

Mamba and related selective SSMs match Transformers on many language tasks while running in linear time, but remain markedly weaker at *associative recall* — retrieving a specific value bound to a key seen earlier in context. Jelassi et al. [1] show this stems from a fundamental capacity constraint: a fixed state cannot losslessly copy arbitrarily long inputs. Yet a capacity bound does not, by itself, explain *behavioral* recall failure at moderate load. The failure could be (a) a **control** problem — the recurrence applies a default recency decay and does not preferentially keep query-relevant items — or (b) a **storage** problem — the state is genuinely out of bits. These have opposite implications: (a) is fixable at inference; (b) is not.

This paper provides evidence for (a) in a regime where it matters. We do *not* modify or retrain the model. Instead we ask: does a pretrained Mamba already expose a handle that lets us trade its default recency policy for a target-preserving one? We find that it does, that the handle is the Δ gate, and — most importantly — that the handle is **localized to a contiguous band of middle layers**. Operating only on that band recovers recall essentially for free in terms of language-modeling quality, the behavior one expects of a dedicated mechanism rather than a diffuse one.

Contributions. (1) A test-time intervention on the Δ gate that controllably preserves a target item in a capacity-pressured SSM, with dose-response, specificity, and bidirectionality controls (§4.1). (2) Localization of this control to **layers 16–23** of Mamba-790m, and a **surgical** result: the 8-layer intervention matches most of the full-model recall gain at perplexity $\times 1.07$ vs. $\times 1.81$ (§4.2). This is our central finding. (3) A framing placing this control on a single axis

with importance-aware KV-cache compression, plus an honest negative result: a literal write-quantization analogue fails, bounding how far the cache analogy carries (§5). We report scope honestly: all positive results are for Mamba-790m on synthetic MQAR; we observe non-replication on Mamba-1.4b and fragility on natural language (§6).

2 Background and Framing

The Δ gate as a retention knob. In a selective SSM the state updates as $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, with $\bar{A}_t = \exp(\Delta_t A)$ and $A < 0$, so $\Delta_t > 0 \Rightarrow \bar{A}_t \in (0, 1)$. The contribution of token i surviving to a query at position T is therefore $\bar{B}_i x_i \cdot \prod_{\tau>i} \bar{A}_\tau$. Two facts follow. First, the default policy is recency-biased: older contributions decay geometrically. Second, the *survival* of an item is governed by the *downstream* decays $\prod_{\tau>i} \bar{A}_\tau$, not by the item’s own write alone. Reducing Δ at positions after a target pushes those $\bar{A}_\tau \rightarrow 1$, slowing the decay the target experiences — a preservation lever requiring no weight change.

Discrete $\leftrightarrow$ continuous importance allocation. The KV-cache compression literature (KIVI [2], KVQuant [3], Zip-Cache [4], and rotation-based quantizers such as Turbo-Quant [5]) implements one principle: spend more bits on more important tokens. This presumes a slotted cache with random access. Mamba has no slots — all items are superimposed in one vector. We therefore treat “importance $\rightarrow$ retained information” as architecture-agnostic and instantiate it in Mamba not as per-token bit allocation but as *differential retention* through the Δ gate. We name this framing Priority-Conditioned Retention (PCR). §5 reports where the analogy holds and breaks.

Table 1: Test-time intervention on Mamba-790m, MQAR $N=32$ (base recall 0.53, base ppl ≈ 232).

Intervention	Recall	Perplexity
All layers, $g=0.8$	0.73	$\times 1.81$
L16–23 only , $g=0.8$	0.67	$\times 1.07$

3 Method

Task. Multi-query associative recall (MQAR) with random single-token key/value words (no semantic priors). A prompt presents N key–value pairs followed by a query key; the model must emit the bound value. Capacity pressure is induced by increasing N ; we query the first pair as the default target.

Intervention. We register forward hooks on each layer’s `dt_proj` and multiply its pre-softplus output by a per-position gain g . Setting $g < 1$ (“other-gain”) at positions downstream of the target’s value reduces their effective decay. A layer mask restricts the intervention to a chosen subset of layers. The model runs on its pure-PyTorch sequential path (fused kernels intentionally not installed) so the Δ pathway is hookable. No parameters are updated.

Model and metrics. `state-spaces/mamba-790m-hf` (48 layers, fp32). We report top-1 recall over 30 random seeds per condition, and perplexity on a fixed held-out passage under the same gain, relative to the unmodified model.

4 Results

4.1 Test-time control

Sweeping other-gain at $N=32$ over all layers yields a clear inverted-U with an optimum at $g=0.8$: recall rises $0.53 \rightarrow 0.73$, then collapses for $g \leq 0.6$ as integration halts. The effect is **target-specific**: applying $g=0.8$ to downstream positions yields 0.73, while the same budget on random positions yields only 0.50 (near base), ruling out a generic “more gain helps” artifact.

4.2 Layer localization and the surgical circuit

Restricting $g=0.8$ to 8-layer bands isolates the responsible depth: recovery concentrates in **L16–23** ($\Delta+0.13$), with a secondary contribution at L24–31 ($+0.07$) and nothing elsewhere. Crucially, the localization is also *cheaper*: Table 1 shows the 8-layer intervention captures most of the recall gain at near-zero language-modeling cost, while the all-layer version pays a large perplexity tax. A targeted edit to the memory circuit is almost free; a global one is not.

5 The Discrete–Continuous Boundary

If retention is one lever, *precision* — the literal KV-quant analogue — is a candidate second lever: quantize the write of low-priority tokens to fewer bits. We tested this by quantizing the per-position convolution output (the token’s written rep-

resentation) to k bits. Single-item precision is not a graceful knob: target recall is flat from $16 \rightarrow 8$ bits, then cliffs ($8\text{-bit } 0.53 \rightarrow 4\text{-bit } 0.03 \rightarrow 0$). Reallocation fails outright: quantizing *distractor* positions to 4 bits drives target recall to 0.00 — it corrupts the shared computation rather than freeing capacity.

This is informative, not merely a null. Because the state is a superposition with no random-access slots, lowering one item’s write precision either does nothing ($\geq 8\text{-bit}$) or injects noise the whole sequence shares ($\leq 4\text{-bit}$). There is no per-item bit budget to reallocate. The importance-allocation principle thus transfers to Mamba *only* through retention (§4.2), not through quantization — a concrete boundary on the KV-cache analogy and an empirical instance of the “no slots” property that distinguishes continuous-state memory from a discrete cache.

6 Limitations and Scope

We state these plainly; they bound the claim and we believe reporting them strengthens it. **Single model:** all positive results are for Mamba-790m; on Mamba-1.4b the same intervention shows no positive circuit at a comparable pressure regime (all layer-band $\Delta \leq 0$ despite ample headroom), so the L16–23 circuit appears model-specific and we do not claim a universal depth. **Synthetic task:** effects are robust on random-token MQAR but fragile on natural-language multi-fact recall, where the intervention can hurt. **Graded priority is not established:** a three-level graded variant appeared monotone but is confounded with query position; we do not claim multi-level allocation. The SSM constraint ($\Delta > 0 \Rightarrow \bar{A} < 1$, shared decay) predicts that truly item-selective allocation is limited, consistent with §5.

7 Related Work

SSM capacity. Jelassi et al. [1] establish the copying/capacity limit; work on primacy/recency documents the default decay policy. We add a controllability result on top of the capacity picture. **KV-cache compression.** KIVI, KVQuant, Zip-Cache and rotation-based quantizers (TurboQuant [5]) realize importance-aware bit allocation on slotted caches; we ask whether the principle survives the move to a slotless continuous state, and find it survives as retention but not as quantization. **Recall and the “what to keep” frontier.** Prompting methods such as Just Read Twice [6] and architectures such as Based, DeltaNet and Titans [7] target the same problem, mostly by training new architectures or memory modules; ours is a training-free, frozen-base alternative. **Mechanistic interpretability.** Localizing a behavior to a contiguous layer band parallels circuit-style findings in Transformers; ours is, to our knowledge, an early such localization of a *controllable* memory mechanism in a pretrained SSM.

8 Conclusion and Future Work

A pretrained Mamba’s associative recall is, in part, a controllable property of a specific middle-layer band, not a fixed

ceiling. A surgical, training-free intervention on layers 16–23 recovers recall at negligible language-modeling cost. The importance-aware compression principle transfers to continuous state through retention but not through quantization. Our proposed next step is to remove the oracle: a small *question-conditioned, auto-localizing* controller trained on a frozen base, evaluated against Belady-MIN retention efficiency under streaming queries, toward a training-free improvement of the recall–capacity frontier of pretrained SSMs.

References

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